Range of a projectile

lecture # 5

If $R = u^2 \sin(2\theta) / g$, with u and g constant, find θ at which R is maximum.

Rmax when $\sin(2\theta)$ is maximum, so $\sin(2\theta) = 1$ $\quad 2\theta = 90°$ $\quad \theta = 45°$

$\theta = 45°$ $\quad R_{max} = u^2 / g$

This is why a level-ground projectile goes farthest at 45°.

If $R = u^2 \sin(2\theta) / g$

u, g, find θ where R is max

$$\max(R) = \max\left(u^2 \cdot \sin(2\theta) / g\right)$$

$$= \frac{u^2}{g}\left(\max(\sin 2\theta)\right.$$

Rough work / extra work

$\max(\sin 2\theta) =$

put $2\theta = \theta_1$

$\max(\sin \theta_1) = 1$

↳ Always

put $\theta_1 = 2\theta$

$\sin 2\theta = \sin\left(\frac{\pi}{2}\right)$

$\theta_1 = \theta_2$

$\Rightarrow$

$\sin(A) = \sin(B)$

1 1

$A = B$

$2\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{4} \Rightarrow \frac{180}{4} = 45^\circ$

Add like sines first

If y = 3 sin θ + 4 sin θ, find ymax.

y = 7 sin θ ymax = 7 × 1 = 7 at θ = 90°

Do not open √(3²+4²) here. Both terms are sine of the same angle. Add the 3 and the 4.

$$y = 7\sin\theta \rightarrow = 1$$

$$\max y = 7 \max(\sin\theta)$$

$$= 7 \times 1$$

A constant sitting next to the sine

Find ymax and ymin.

$y = 3 + 2\sin\theta$ ymax $= 3 + 2(1) = 5$ ymin $= 3 + 2(-1) = 1$

$y = 3 + 4\sin\theta$ ymax $= 7$ ymin $= 3 - 4 = -1$

$y = 3\sin\theta - 4$ ymax $= 3 - 4 = -1$ ymin $= -3 - 4 = -7$

$3 + 2\sin\theta \rightarrow 5$ and 1 $3 + 4\sin\theta \rightarrow 7$ and -1 $3\sin\theta - 4 \rightarrow -1$ and -7

-1

-7

Rough work / extra work

$y = \sin\theta + \cos\theta$

$\max(y) = ?$

If $\max(y) = \max(\sin\theta) + \max(\cos\theta)$ # It will fail

θ = same both a & b

$\max(\sin\theta) \Rightarrow \theta = 90 = 1$ → (1)

& At $\theta = 90$

$\cos(90) = 0$ → (2)

Rough work / extra work

by Eqn 1 & 2

$\max(y) = 1 + 0 = \underline{1} \rightarrow$

$\sin(\theta) \Rightarrow$ at $\theta = 45^\circ = \frac{1}{\sqrt{2}}$

$\cos(45) = \frac{1}{\sqrt{2}}$

$y_{\text{at } \theta = 45} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \rightarrow \frac{\sqrt{2} \times \sqrt{2}}{\sqrt{2}} = \sqrt{2}$

$= 1.414$

much greater 1 (at $\theta = 90$)

Rough work / extra work

$$\max\left(\frac{var1}{b} \quad \frac{var2}{b}\right) \neq \max(var1 + var2)$$

(not always)

If b both positive then Ib can be true but not always when we have θ

case 1) $\theta = 45 \Rightarrow$ True (above Example)

but $\theta = 60^\circ \rightarrow y = \sin 60 + \cos(60)$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} \Rightarrow \frac{1.73}{2} \Rightarrow \frac{0.86 + 0.5}{}$$

$\Rightarrow$ 1.36 > 1

Rough work / extra work

$(\sin\theta + \cos\theta) \neq 1$

a, b

$\sqrt{a^2 + b^2}$ (max)

$\sqrt{\sin^2\theta + \cos^2\theta}$

$\sqrt{1}$

(1.41)

$\sqrt{a^2 + 3^2}$

$\because \cos\theta = \sin(90-\theta)$

Rough work / extra work

$\frac{1}{1.41} =$

$(\sin\theta + \cos\theta)$

$\theta = 90 \rightarrow \sin\theta = 1$

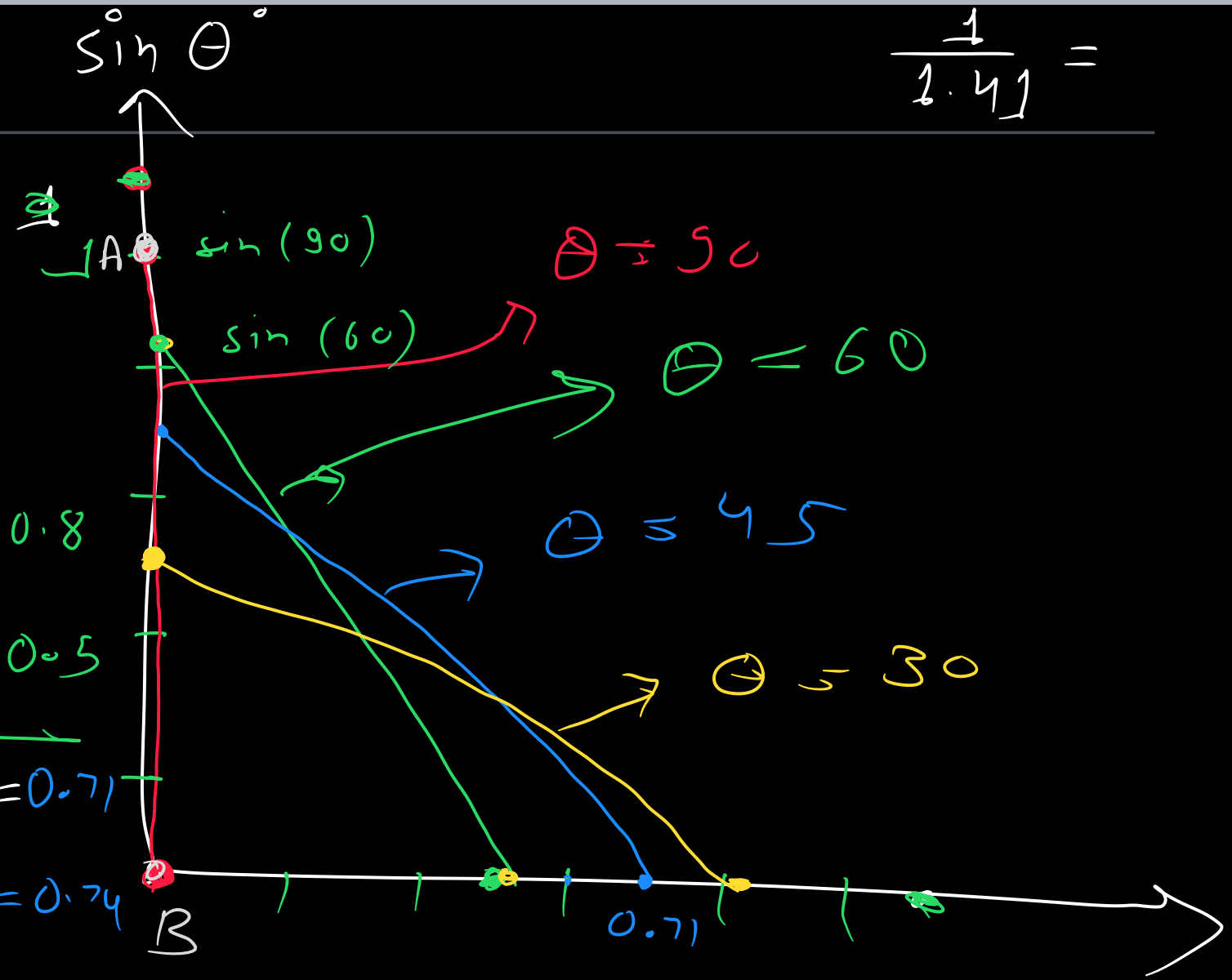
$\cos = 0$

$\theta = 60^\circ \rightarrow \sin 60 = \frac{\sqrt{3}}{2} = 0.8$

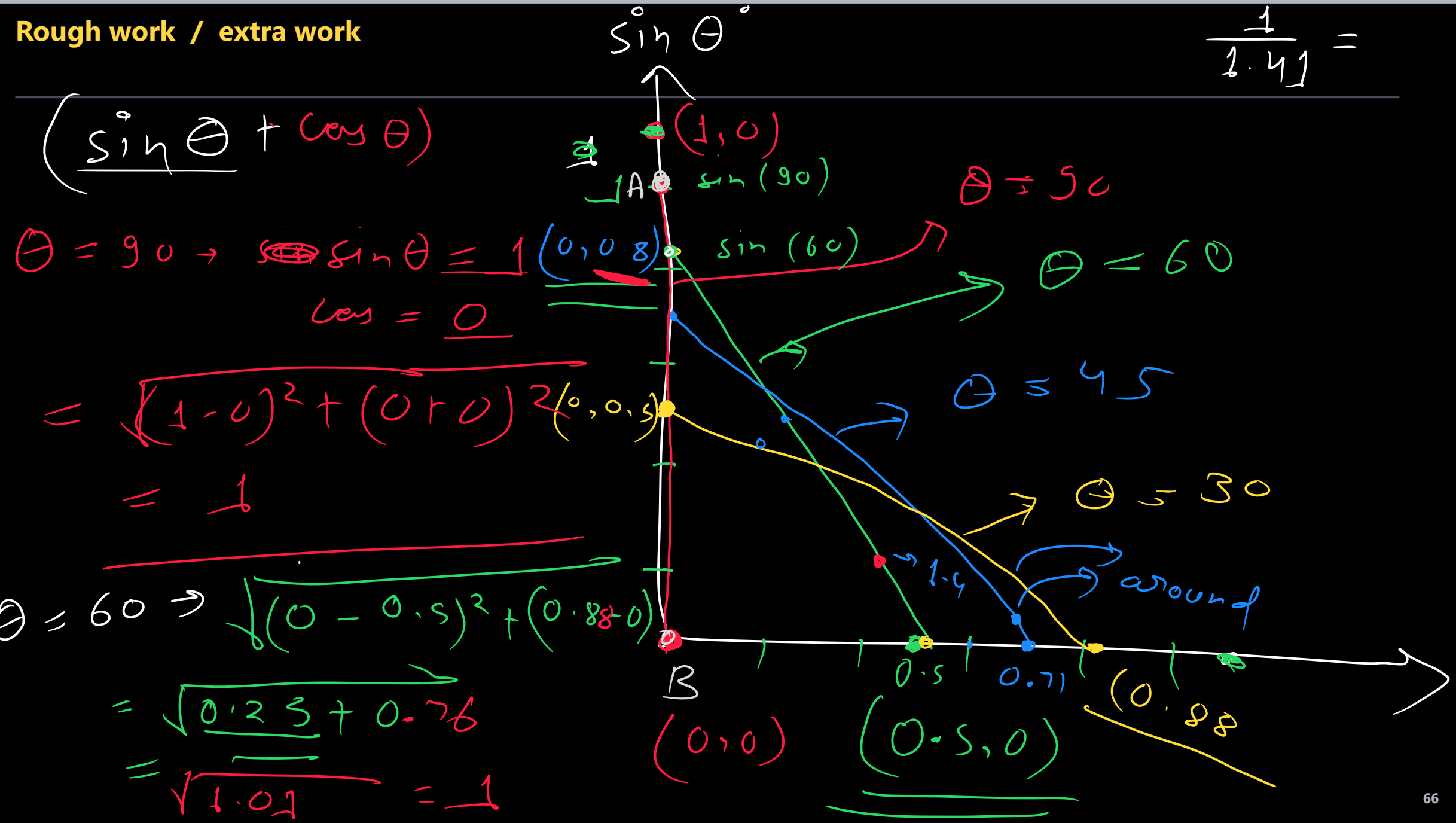
$\cos 60 = 1/2 = 0.5$

$\theta = 45 \rightarrow \sin 45 = 1/\sqrt{3} = 0.71$

$\cos 45 = 1/\sqrt{2} = 0.74$



Rough work / extra work



Rough work / extra work

θ = 45

↳ 1.41

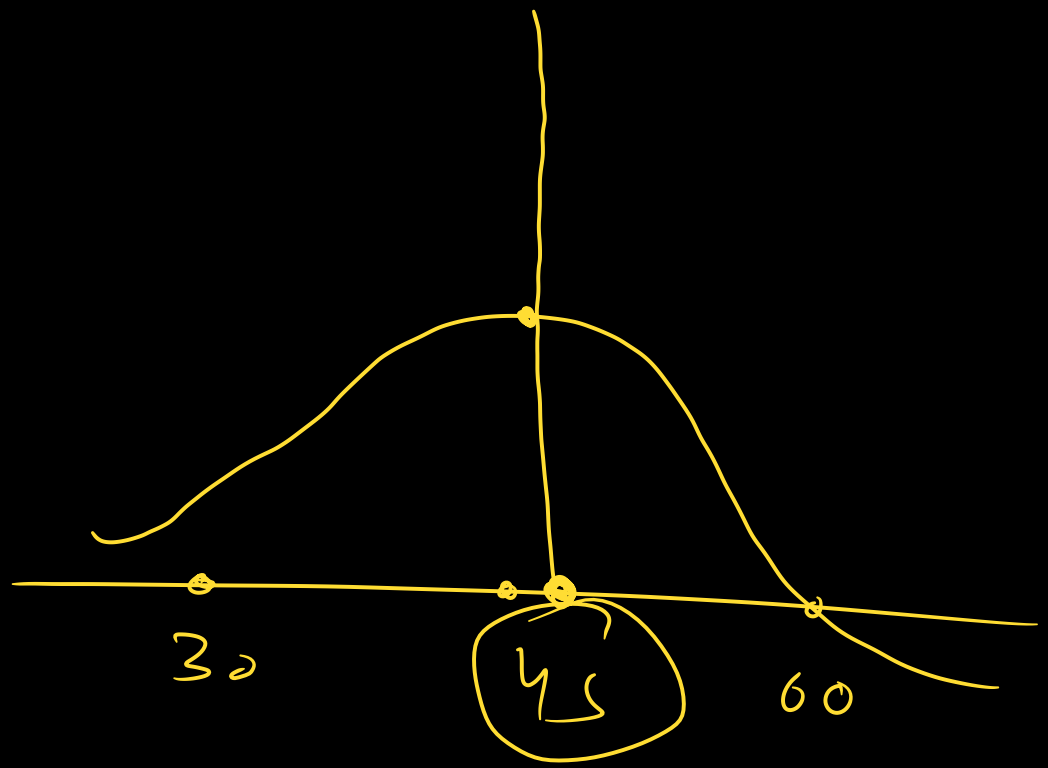
↳

θ = 30 ≤ 1.1 →

θ = 60 → 1.1

θ = 45 → 1.4

0 30 ≤ 1.1

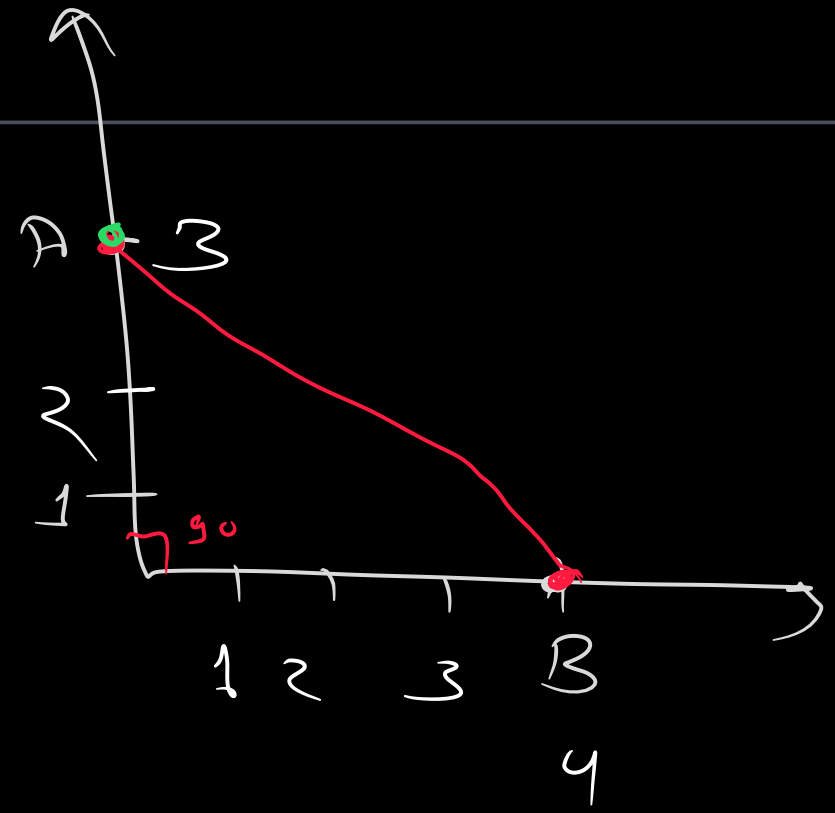


Rough work / extra work

Distance Plane

$AB = 3^2 + 4^2$

$= 25 = 5$



Rough work / extra work

$$\frac{1}{1.4} = \frac{10}{14}$$

28
42
56
70
84
98
112
126

$$\frac{98}{14} \quad \frac{0.71}{1}$$

30

Rough work / extra work

$\sqrt{2} = 1.414$

$\sqrt{3} = 1.73$

$\sqrt{4} = 2$

$\sqrt{5} = = 2.256 \rightarrow (2.25)^2 \Rightarrow 5$

$\sqrt{6} \rightarrow \sqrt{2} \times \sqrt{3} = 1.41 \times 1.73 = 2.4393 = 2.44$ (2 decimal)

$\sqrt{8} = ? = \sqrt{2 \times 2 \times 2} = 2\sqrt{2} = 2 \times 1.4 = 2.8$

$\sqrt{9} = 3$

Rough work / extra work

1.41 + 1.7

1.41
1.70

3.11

1.73
1.41

When sine and cosine are added

sin θ and cos θ do not peak together. The combined amplitude is the hypotenuse of a and b.

$y = a \sin\theta + b \cos\theta$

$$y_{max} = +\sqrt{(a^2 + b^2)} \qquad y_{min} = -\sqrt{(a^2 + b^2)}$$

Do not write ymax = a + b. That would need sin = 1 and cos = 1 at the same instant, which never happens.

$y = 3 \sin \theta + 4 \cos \theta$

If $y = 3 \sin \theta + 4 \cos \theta$, find ymax and ymin.

$y_{max} = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = +5$

$y_{min} = -5$

$y_{max} = +5 \qquad y_{min} = -5$

Minimum magnitude of a fraction

H.W

do it ?
Google

If y = 4 / (3 sin θ + 2 cos θ), find the minimum magnitude of y.

4/10 = 0.4. 4/100 = 0.04. A larger denominator makes a smaller positive fraction.

ymin = 4 / (3 sin θ + 2 cos θ)max = 4 / √(3² + 2²) = 4 / √13

minimum magnitude = 4 / √13

Five max–min — every line

Find maximum and minimum of each.

	y	ymax	ymin
(a)	$4\sin\theta + 3\cos\theta$	$\sqrt{(16+9)} = 5$	-5
(b)	$2\sin\theta - 3\cos\theta$	$\sqrt{(4+9)} = \sqrt{13}$	$-\sqrt{13}$
(c)	$1 + 5\sin\theta$	$1+5 = 6$	$1-5 = -4$
(d)	$1 - 2\sin\theta$	$1-2(-1) = 3$	$1-2(1) = -1$
(e)	$5\sin\theta - 12\cos\theta$	$\sqrt{(25+144)} = 13$	-13